Ngoc Thanh Nguyen · Tokuro Matsuo ·
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Recent Challenges in Intelligent Information and Database Systems

17th Asian Conference, ACIIDS 2025
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Knowledge Graph-Based Legal Query System with LLM and Retrieval Augmented Generation

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Abstract Nowadays, the demand for legal queries is substantial. Therefore, systems must be capable of accurately analyzing the context of queries and retrieving relevant legal regulations. This study introduces a novel intelligent legal query system designed to enhance legal information retrieval. By combining the knowledge base, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs), the system effectively extracts keywords and intricate relationships from legal regulations. RAG is then utilized to retrieve relevant information from extensive databases, while LLMs process and generate comprehensive, human-readable responses. This integrated approach surpasses traditional methods in terms of speed and accuracy, providing a significant advancement in legal information access and utilization.

Keywords: Knowledge base · Knowledge graph · Legal document · Large language model · Retrieval-augmented generation

1 Introduction

In a rapidly evolving society, the role of law is essential in maintaining order and justice. However, accessing and applying legal information effectively remains a significant challenge. It encompasses principles and arguments that guide the application of legal regulations to real-world situations [20]. The process of understanding and gathering legal information involves searching, and analyzing legal documents to effectively apply them in practice [6]. This enables individuals and organizations to quickly access legal regulations, ensuring precise and efficient compliance. Nevertheless, this process faces many challenges due to the

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large volume of documents, complex specialized language, and a lack of intelligent automation tools [13]. These challenges underscore the need for innovative approaches to enable more accurate legal information retrieval.

Recent studies have highlighted the significant potential of RAG and LLMs in enhancing the efficiency of legal information retrieval [10,18]. RAG facilitates quick and accurate access to vast legal data repositories, while LLMs analyze and generate semantically meaningful responses based on specific contexts [7]. Despite these advancements, challenges remain in optimizing computational resources and ensuring accuracy from diverse, unstructured sources [3].

This study proposes an intelligent legal query system that integrates knowledge graphs, RAG, and LLMs to optimize the process of legal information retrieval based on ontology Legal-Onto [8,14]. Knowledge graphs are employed to extract keywords and concept triples while elucidating complex relationships between legal regulations. Subsequently, RAG retrieves information from large databases, and LLMs analyze and generate semantically clear responses based on the retrieved data. This integrated approach leverages the strengths of each component to address the limitations of traditional methods, offering a novel solution to legal information retrieval. Experimental results demonstrate its superior speed and accuracy in handling complex legal regulations.

The next section reviews related studies on RAG, LLMs, and knowledge graphs in legal queries, offering insights into current advancements and challenges. Section 3 elaborates on the proposed method, including the integration of RAG and LLMs with knowledge graphs to optimize the query process. Section 4 describes the experimental process, comparing the effectiveness of the new system with traditional query methods. Finally, Sect. 5 presents conclusions and suggestions for future research, focusing on expanding applications and enhancing automation capabilities in other legal domains.

2 Related Work

In the legal domain, Knowledge Graph (KG) has become an important research direction to enhance the efficiency of information retrieval [1]. Numerous studies have leveraged knowledge extracted from legal texts to help users access information more quickly and accurately [16]. An ontology-based approach is applied to construct linked data from approximately 325,000 Vietnamese legal documents, modeling legal knowledge as a conceptual graph [12,13]. However, this approach faces challenges in updating semantics when regulatory changes occur, which limits the effectiveness of queries.

The study in [2] focuses on enhancing the resolution of complex linguistic tasks through fine-tuning LLMs, RAG, and knowledge graphs. This approach integrates fine-tuning LLMs with RAG techniques and knowledge graphs to optimize the process of understanding and generating accurate responses. The results show that this method significantly improves performance in complex linguistic tasks. However, challenges remain when applying it to situations with intricate semantics, and further improvements are needed to achieve higher accuracy in such cases.

The CBR-RAG system is combined with Case-Based Reasoning (CBR) with RAG and LLMs to improve accuracy in legal question answering [18]. This system uses relevant legal cases as contextual input, enhancing the semantic meaning of the responses. However, the system still faces challenges in handling large and complex legal datasets, affecting retrieval speed and scalability.

Although significant progress has been made in legal information retrieval, current systems still face several limitations when handling large volumes of legal data, particularly in Vietnam, where there are thousands of legal documents across various levels. In addition, existing systems lack the ability to automatically update and maintain meaning when legal regulations change. The ability to retrieve accurate information from complex and heterogeneous legal sources remains limited [3].

The proposed method combines knowledge graphs with RAG and LLMs to improve legal information retrieval and predict links between legal concepts. It automatically updates meanings when laws change and deeply analyzes text to handle complex queries, ensuring more accurate, context-based results.

3 The Integration of RAG and Knowledge Graphs for Enhanced Legal Querying

3.1 Knowledge-Based Structure

To optimally represent legal knowledge, the ontology model, including concepts and their relationships, suitable for expressing knowledge in Vietnamese legal documents, is applied [17]. Ontology Rela-model is a knowledge base that aims to improve reasoning and problem-solving capabilities in specialized fields [9]. This model consists of three main components: concepts (objects), relations, and rules for information inferring. From those characteristics, legal knowledge modeling on Vietnamese traffic laws is implemented.

Definition 3.1: [6] The Ontology Legal-Onto model is structured as follows:

$$K = (C, R, \text{Rules}) + (\text{Conc}, \text{Rel})$$

There are two main parts in the above structure:

1. **Rela-model Structure (C, R, Rules)**
   - C: The set of concepts, where each concept has been refined to represent legal content. Each concept has the following structure:

     $$(\text{Name}, \text{Meaning}, \text{Attrs}, \text{Keyphrases}, \text{Similar})$$

     where, **Name** is the name of the concept. **Meaning** is the legal content of the concept, defined by legal documents. **Attrs** is a set of attributes of the concept. **Keyphrases** is s set of keywords related to the concept. **Similar** is a list of different representations of the concept.

- R: The set of relations between concepts in the system, describing how these concepts are connected or dependent on each other, reflecting the content of legal documents. Each relation has the structure:

$$(\text{Name}, \text{Meaning}, \text{ConckeyS}, \text{ConckeyO}, \text{Prop}, \text{Keywords})$$

  where, **Name** and **Meaning** are the name and meaning of the relation. **ConckeyS** and **ConckeyO** are the concepts serving as the subject and object of the relation (resp.). **Prop** is a set of properties.
- **Rules**: The set of logical rules used to infer specific knowledge, applied to the components of the knowledge representation.

2. **Knowledge Graph (Conc, Rel)**
   - **Conc**: The set of terms that reflect concepts and specialized terminology found in legal documents.
   - **Rel**: The set of semantic links (directed arcs), which are directed edges representing relationships between concepts in the legal domain.

3.2 Building Knowledge Graphs

Knowledge graphs utilize a graph-based data model to capture, integrate, manage, and exploit knowledge from multiple large-scale data sources [21]. Knowledge graphs use a structure based on semantic triples (*subject, predicate, object*) to represent, integrate, and extract knowledge from diverse data sources [5]. In this structure: The *subject* represents the entity, The *predicate* represents the relations, and The *object* represents the target entity.

Each triple corresponds to an edge in the graph, linking nodes that represent the subject and object, enabling the modeling of complex relationships between entities. Integrating the triple structure into the *Legal-Onto* model requires a mapping method [13].

- For each $c \in C$ corresponding to an entity in the semantic triple, c will represent a term equivalent to the subject or object in the triple.
- For each $r \in R$ corresponding to a relationship in the semantic triple, r will represent a term equivalent to the predicate in the triple.

Pos-tagging method using VnCoreNLP - a tool to support Vietnamese language processing. When applying VnCoreNLP to legal documents, each component in the sentence will be specifically labeled according to its part of speech, thereby creating a clear grammatical structure for the document and supporting further analysis steps. Based on the results of the POS-Tagging method, the output is organized into a tree structure where each node represents a token containing information about its part-of-speech (posTag) and dependency label (depLabel). The posTags, such as N (Noun), P (Pronoun), and V (Verb), along with dependency labels like sub (Subject) and dob (Direct Object), are used

Algorithm 1. Triplet Extraction Algorithm

Input: A natural language text.
Output: List of triplets in this text.
Algorithm:
text ← Preprocess a text
list_tokens ← annotate the text with VnCoreNLP
list_triplets ← {}
for each token of *list_tokens* **do**
  if posTag of token is Verb **then**
    Insert $\{\emptyset, token, \emptyset\}$ into *list_triplets*
  end if
  if posTag of token is Noun **then**
    triple ← Search the triple of *list_triplets* which is header of token
    if depLabel of token is Subject **then**
      first element of *triple* ← *token*
    end if
    if depLabel of token is Object **then**
      third element of *triple* ← *token*
    end if
  end if
  if token is a Conjunction word **then**
    main_token ← Search a main token of reference sentence
    refer_triple ← Search a triple of *main_token*
    new_triple ← Define a new triple with the same token as *refer_triple* but replace the token word
    Insert *new_triple* into *list_triplets*
  end if
end for
Return *list_triplets*

to identify potential entities that form semantic triplets. The triplet extraction algorithm is as follows:

Figure 1 illustrates the processing of Algorithm 1 for extraction triplets in legal documents. During the construction of the knowledge graph, relationships with similar or closely related meanings are grouped into a unified relationship to ensure consistency and simplify the graph's structure. Additionally, duplicate semantic triplets are eliminated to optimize performance. These duplicates often arise from data collected from multiple sources or due to repetitive entities and relationships generated during the automatic labeling process [15].

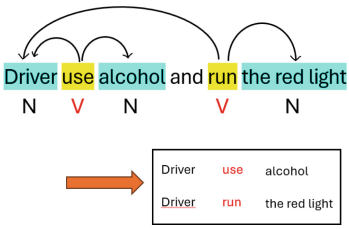


Fig. 1. Triplet extraction from text.

The result of this process is a set of semantic triplets representing legal knowledge extracted from legal documents. Each triplet is associated with a list of IDs corresponding to the legal documents, clearly identifying the source and context of the knowledge. This allows the system to utilize triplets to address legal questions.

3.3 Design of the RAG Query Method

3.3.1 Theoretical Foundation An effective solution to mitigate the issue of "hallucinations" in language models is the use of RAG [19]. "Hallucinations" refer to the phenomenon where a language model generates inaccurate, misleading, or contextually irrelevant information, making the response appear reliable while, in fact, being incorrect [11]. RAG combines two main stages:

1. **Information retrieval from the knowledge base**: Based on the query, the system retrieves relevant information from the knowledge base.
2. **Answer generation**: The LLM utilizes the retrieved information to generate an appropriate response, minimizing inaccurate or contextually irrelevant information.

This approach uses pre-trained large language models like BART [11] to generate text. It integrates knowledge graphs with GPT-4o through RAG to extract legal knowledge from structured and unstructured data sources, specifically "thuvienphapluat.vn.". The knowledge graph provides background in semantic triples, while GPT-4o process natural language and generates responses based on this information. This method ensures accuracy and enhances logical consistency, enabling quick access to legal infomation.

3.3.2 Query Method on the Knowledge Domain The query system combines knowledge graphs and LLM to process and respond to complex queries. The process involves the following steps:

Step 1: Knowledge Extraction from Knowledge Graphs

- From a natural language query, the system extracts entities and relationships, representing them as semantic triples [15].
- These triples are matched and mapped onto the knowledge graph to identify relevant subgraphs.
- The result is a set of legal documents aligned with the query.

Step 2: Integration of Knowledge Graph Data into LLMs

- LLMs receive input consisting of selected legal documents from the knowledge graph, alongside the original query.
- Specifically designed prompts guide GPT-4o to generate responses based on the retrieved data.

Step 3: Answer Generation

- The LLM uses its semantic and contextual understanding capabilities to provide a suitable answer.
- The information from the knowledge graph ensures that the response is based on reliable legal grounds.

Example 3.1: **Query**: "Tôi chạy xe khi đã uống rượu bia và vượt đèn đỏ. Tôi bị khuyết tật. (English: I drove car after drinking alcohol and ran a red light. I am disabled.)"

Extracted Triplets:

- Tôi (I) - chạy (drove) - xe (car) (*Set Document Id: A*).
- Tôi (I) - uống (drinking) - rượu_bia (alcohol) (*Set Document Id: B*).
- Tôi (I) - vượt (ran) - đèn_đỏ (a red light) (*Set Document Id: C*).
- Tôi (I) - bị (am) - khuyết_tật (disabled) (*Set Document Id: D*).

From these triples, the system identifies relevant legal documents. Using heuristics, it determines the most pertinent set of documents. During the information retrieval phase, the system employs a query model to search for and filter the most relevant information from the knowledge base. Heuristic methods are implemented to identify and extract entities as well as relationships that align with the given query. In the practice, the initial set of retrieved documents may lack a clear commonality. To extract the most relevant and cohesive documents, the heuristic method is applied based on pairwise intersections of legal document sets as follows:

- **If the intersection is non-empty**: The intersected set constitutes the critical documents necessary to answer the query.
- **If the intersection is empty**: This indicates that one or both sets may be relevant to the query but operate independently with no direct overlap.

Let the *RelatedIds* be a list of document IDs related to the query, defined by the formula:

$$RelatedIds = \bigcup_{1 \le i < j \le n} (S_i \cap S_j) \cup \{S_u \mid \forall v \in [1, n] \setminus \{u\},\ S_u \cap S_v = \varnothing\}$$

After gathering sufficient information from the knowledge base, the system proceeds to the answer generation phase. At this stage, the LLM utilizes the retrieved data to generate answers or provide corresponding legal conclusions. The generated results are based on information from the knowledge base and are refined to ensure accuracy, reliability, and relevance to the query.

The application of the RAG model in legal knowledge querying not only enhances the efficiency of information retrieval but also significantly improves the accuracy and speed of providing answers. Consequently, the system enables users to access legal information more quickly, effectively, and accurately.

3.3.3 The Architecture of Integrating the Legal Retrieval and LLM Based on Knowledge Graph When there is knowledge directly related to the query, the comprehension capability of LLMs becomes crucial in addressing the user's need for a complete and accurate answer. LLMs can analyze the query, leverage relevant knowledge, and synthesize a response that aligns with the user's expectations. However, for optimal performance, the model requires a well-designed and specific prompt to correctly interpret the query and generate an appropriate response. A good prompt must adhere closely to the context of the query, minimizing unnecessary words or ambiguity while maintaining clarity. The structure of the prompt should follow a logical sequence, presenting the information in a way that the model can easily process.

Prompt Design

- **System Prompt:** You are a good lawyer. You help users answer questions about Vietnamese law. You answer accurately about legal documents.
- **User Prompt:** Vietnamese Legal Document: {documents}. Based on the relevant legal documents above, answer the below question:
  Question: {question}
  Your answer in Vietnamese language.

  Example 3.2: The prompt:
- **Input:** Vietnamese Legal Document: Decree 100/2019/ND-CP about Administrative penalties for road traffic and rail transport offences.
  Based on the relevant legal documents above, Question: "Hành vi phơi thóc, lúa, rơm rạ trên đường bộ có bị xử phạt không?" (English: *Is it punishable to dry paddy, rice, straw, or hay on public roads?*)
- **Output:**"Hành vi phơi thóc, lúa, rơm rạ trên đường bộ sẽ bị xử phạt theo Nghị Định 100/2019/ND-CP, cụ thể tại Điều 12, Khoản 1, Điểm b. Theo đó, phạt tiền từ 100.000 đồng đến 200.000 đồng đối với cá nhân thực hiện hành vi này. Nếu có tổ chức thực hiện hành vi phơi thóc, lúa, rơm rạ trên đường bộ thì mức phạt sẽ từ 200.000 đồng đến 400.000 đồng. (English: *The act of drying paddy, rice, straw, or hay on public roads is subject to a fine according to Decree 100/2019/ND-CP, specifically Article 12, Clause 1, Point b. Accordingly, individuals who commit this act will be fined from 100,000 to 200,000 VND. If an organization is involved in drying paddy, rice, straw, or hay on public roads, the fine will range from 200,000 to 400,000 VND.*)

The Architecture of Legal Retrieval System

The system architecture integrates multiple components to efficiently process legal queries. The core employs a pre-trained large language model (LLM) alongside a retrieval model and a knowledge graph (KG), as Fig. 2. User queries are submitted through the user interface analyzed by the query analysis engine. The

system relies on semantic triplets extracted from queries to perform knowledge graph mapping This structured approach enhances the system's ability to process and analyze complex legal data. Figure 2 displays the architecture for legal information retrieval integrating KGs, RAG, and LLMs.

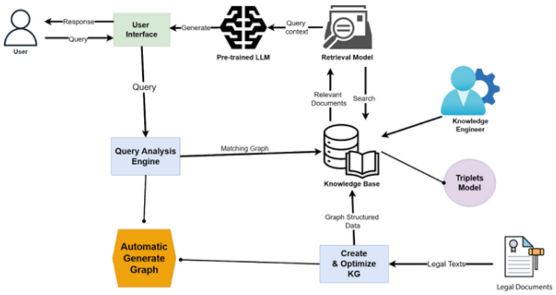


Fig. 2. The architecture for legal information retrieval.

The process begins when the user submits a natural language query through the interface. The query is analyzed to extract semantic triplets, which are mapped to the knowledge graph to find relevant legal documents. These documents, retrieved via the retrieval model, are passed to the LLM for deeper semantic analysis. The LLM generates a response based on the retrieved data, ensuring accuracy and contextual relevance, as shown in Fig. 3. Finally, the system refines the output to provide a complete and precise answer to the user. This streamlined process ensures quick and accurate access to legal information. Figure 3 presents the processing flow of the legal query system.

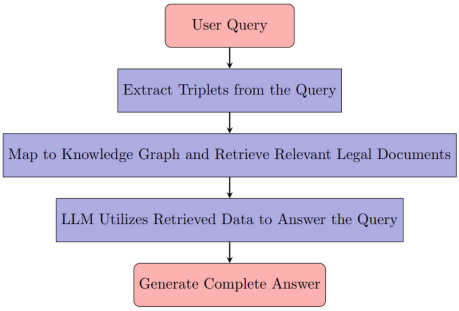


Fig. 3. The processing of the legal query system.

The integration of RAG, LLMs, and knowledge graphs presents a comprehensive solution to optimize the legal query process. By leveraging the strengths of each component, this approach addresses key challenges in legal information

retrieval, including handling complex queries, ensuring accuracy, and adapting to changes in legal regulations. It also enhances the analysis of semantic relationships in legal contexts.

4 Experimentation and Results

4.1 Experimentation

To evaluate the system's accuracy, approximately 200 question-and-answer pairs related to legal matters were collected from a Vietnamese legal forum "thuvienphapluat.vn.". These questions are natural sentences written by humans. Each question is paired with an answer provided by professional lawyers. For each question, the system is queried to compare its results with the reference answers.

The system demonstrated high accuracy in retrieving relevant legal provisions. However, errors occurred in answer generation due to the retrieval of too many unrelated provisions, causing the LLM to select less relevant results. The detailed evaluation results are summarized in Table 1, which presents the system's performance in two areas: (1) the accuracy of answer generation and (2) the accuracy of legal provision retrieval. While retrieval accuracy was notably high, the accuracy of answer generation revealed areas for improvement, as errors often stemmed from the inclusion of irrelevant provisions during retrieval.

Table 1. Results of the Legal Query System Evaluation

Query Type	Quantity	Answer Generation		Legal Retrieval	
		Correct	Percent	Correct	Percent
Concepts	40	28	70.00%	33	82.50%
Q&A	157	111	70.70%	127	80.89%
Total	**197**	**139**	**70.56%**	**160**	**81.22%**

4.2 Comparison with Other LLMs

Currently, large language models such as ChatGPT, Gemini, and Copilot have emerged as versatile chatbot systems capable of handling various tasks, including querying knowledge in the field of legal consultation. ChatGPT, based on the GPT-4 model by OpenAI, excels in natural language processing tasks. Gemini, built on the Gemini Pro model by Google, specializes in combining natural language understanding with programming logic. Table 2 shows the comparison results based these criteria.

The results of the proposed method are better than other LLMs. The other LLMs utilize existing online content without verifying the validity of legal provisions. As a result, they integrate information from both outdated and updated laws, leading to inaccurate responses. The proposed system is built with the ability to update the adjusted laws based on its knowledge base, so its performance is more effective than others.

Table 2. Comparison of Legal Query System with other LLMs

Systems	Quantity	Correct	Percent
ChatGPT	197	60	34.52%
Copilot	197	122	61.93%
Gemini	197	131	66.50%
Proposed sytem	197	160	81.22%

5 Conclusion

The legal query system, combining RAG, LLMs, and knowledge graphs, effectively improves legal information retrieval and processing. It enhances accuracy in finding relevant documents and hidden links while generating contextually appropriate answers. This system overcomes many limitations of traditional methods, especially in handling complex queries and updating semantics automatically when laws change.

Future works will focus on improving the system's semantic processing to better analyze complex relationships in legal documents, reduce context-related errors, and uncover hidden legal issues [4]. The system will also be expanded to support multilingual queries, enabling broader applicability on a global scale, and automation will be improved through advanced machine learning methods to dynamically update knowledge graphs as legal regulations evolve.

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